Shreyas G Shet

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Education

BMS Institute of Technology & Management Bengaluru, IN

Bachelor of Engineering, Computer Science and Engineering *2023 — 2027*

- CGPA: 8.72
- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Digital Design and Computer Organization, Database Management Systems, Theory of Computation, Compiler Design, Computer Networks, Cloud Computing, Machine Learning, Microprocessors, Linear Algebra, Discrete Mathematics, Multivariable & Single Variable Calculus

Projects

Custom Memory Allocator GitHub

- Designed and implemented a heap allocator in C from scratch, replicating core mechanics of production allocators like glibc's malloc: explicit free lists, first-fit search, block splitting, forward/backward coalescing, and 16-byte aligned allocation with boundary-tag headers.
- Modeled sbrk-style heap growth on a simulated process heap to mirror real Unix memory management, so the allocator's logic maps directly onto how an OS actually grows and reclaims a process's address space.
- Achieved 91.2% peak heap utilization and 28.6% free-block reuse under controlled workload testing, by instrumenting the allocator to track allocation attempts, splits, and coalescing events, then tuning the splitting/coalescing logic against that data.

Production Task Manager API GitHub

- Built a task management system in Java like the back-end of a to-do list app that lets you create, update, and organize tasks, and can talk to other apps or websites that want to show or manage that list.
- Migrated a Task Management REST API from a raw com.sun.net.httpserver implementation to Spring Boot, replacing manual routing and HTTP method dispatch with `@RestController` endpoints and declarative request mapping, eliminated hand-written routing/dispatch logic, making endpoints easier to extend and less error-prone as the API grew..
- Added centralized error handling via `@ControllerAdvice/@ExceptionHandler`, returning structured, consistent error responses (400/404/500) in place of the original manual status-code logic.
- Containerize the Spring Boot service with Docker, using multi-stage builds to minimize image size and cleanly separate build-time and runtime dependencies.
- Set up a CI/CD pipeline (GitHub Actions) to automate build, test, and container image publishing on every push, cutting down manual deployment steps.

Mini Inference Engine GitHub

- Implemented the GPT-2 forward pass from scratch as plain functions (layer norm, causal multi-head attention, MLP, pre-LN transformer block), with no model-framework classes or library model code.
- Verified correctness against HuggingFace's reference implementation: next-token predictions (argmax) matched at every position across 4 test prompts, with a max absolute logit difference of 1.2e-4.
- Extending the engine with a KV cache and continuous batching, to be benchmarked against static batching on tokens/sec, p95 time-to-first-token, and memory utilization.

Learning-Based Compiler Optimization Pass Selection using Agentic Systems GitHub

- Designed a first-principles methodology for compiler pass-ordering research: identified optimization opportunities in LLVM IR by hand, mapped them to specific LLVM passes, and empirically tested sequencing effects across a 6-pass pipeline (mem2reg, instcombine, simplifycfg, reassociate, newgvn, gvn-hoist).
- Laid the technical foundation for an agentic compiler-optimization system that observes program IR state, selects/applies LLVM passes, and adapts based on newly exposed opportunities — motivated by the research question of whether learned pass sequences can beat static -O2/-O3 pipelines.

Skills

- **Programming Languages**: Python, C, C++, Java, Bash
- **Tooling**: Git, UNIX/Linux, NGINX, Maven, Docker, GitHub Actions, PostgreSQL
- **Libraries & Frameworks**: Spring Boot, NumPy, PyTorch, HuggingFace Transformers